

6. Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{cov}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$

To test the null hypothesis $H_0: \beta_2 = 0$, we compute the t-statistic:

$$t = \frac{b_2}{se(b_2)}$$

The standard error is obtained from the covariance matrix as $se(b_2) = \sqrt{4} = 2$, $t = \frac{3}{2} = 1.5$. With

$n = 63$ and $k = 3$, the degrees of freedom are 60.

At the 5% significance level (two-tailed), the critical value is approximately 2.00. Since $|t| = 1.5 < 2.00$, we fail to reject the null hypothesis.

Therefore, β_2 is not statistically significantly different from zero at the 5% level.

b. $\beta_1 + 2\beta_2 = 5$

To test the null hypothesis $H_0: \beta_1 + 2\beta_2 = 5$, we first compute the estimated value:

$$b_1 + 2b_2 = 2 + 2(3) = 8$$

The variance of this linear combination is:

$$\begin{aligned} Var(b_1 + 2b_2) &= Var(b_1) + 4Var(b_2) + 2 \times 2Cov(b_1, b_2) \\ &= 3 + 4 \times 4 + 2 \times 2 \times (-2) = 11 \end{aligned}$$

Thus, the standard error is $\sqrt{11}$.

The t-statistic is:

$$t = \frac{(b_1 + 2b_2) - 5}{se(b_1 + 2b_2)} = \frac{8 - 5}{\sqrt{11}} \approx 0.9045$$

At the 5% significance level, the critical value is approximately 2. Since $|t| < 2$, we fail to reject the null hypothesis.

c. $\beta_1 - \beta_2 + \beta_3 = 4$

To test the null hypothesis $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, we first compute:

$$b_1 - b_2 + b_3 = 2 - 3 - 1 = -2$$

The variance of this linear combination is:

$$\begin{aligned} Var(b_1 - b_2 + b_3) &= Var(b_1) + Var(b_2) + Var(b_3) - 2Cov(b_1, b_2) + 2Cov(b_1, b_3) - 2Cov(b_2, b_3) \\ &= 3 + 4 + 3 - 2 \times (-2) + 2 \times 1 - 2 \times 0 = 16 \end{aligned}$$

Thus, the standard error is $\sqrt{16} = 4$.

The t-statistic is:

$$t = \frac{(b_1 - b_2 + b_3) - 4}{se(b_1 - b_2 + b_3)} = \frac{-2 - 4}{4} = -1.5$$

At the 5% significance level, the critical value is approximately 2. Since $|t| < 2$, we fail to reject the null hypothesis.